



gemma-3-12b-it

Text Generation • Google

@cf/google/gemma-3-12b-it

Gemma 3 models are well-suited for a variety of text generation and image understanding tasks, including question answering, summarization, and reasoning.

Gemma 3 models are multimodal, handling text and image input and generating text output, with a large, 128K context window, multilingual support in over 140 languages, and is available in more sizes than previous versions.

Model Info

Context 80,000 tokens

Window [↗](#)

Model Info

LoRA	Yes
Unit Pricing	\$0.35 per M input tokens, \$0.56 per M output tokens

Playground

Try out this model with Workers AI LLM Playground. It does not require any setup or authentication and an instant way to preview and test a model directly in the browser.

Launch the LLM Playground

Usage

✓ Worker - Streaming

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const stream = await  
env.AI.run("@cf/google/gemma-3-12b-it", {  
      messages,  
      stream: true,  
    });
```

```
return new Response(stream, {  
  headers: { "content-type":  
"text/event-stream" },  
  });  
},  
} satisfies ExportedHandler<Env>;
```

▼ Worker

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const response = await  
env.AI.run("@cf/google/gemma-3-12b-it", {  
  messages });  
  
    return Response.json(response);  
  },  
} satisfies ExportedHandler<Env>;
```

▼ Python

Python

```
import os
import requests

ACCOUNT_ID = "your-account-id"
AUTH_TOKEN = os.environ.get("CLOUDFLARE_AUTH_TOKEN")

prompt = "Tell me all about PEP-8"
response = requests.post(
    f"https://api.cloudflare.com/client/v4/accounts/{ACCOUNT_ID}/workers-ai/generate",
    headers={"Authorization": f"Bearer {AUTH_TOKEN}"},
    json={
        "messages": [
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt}
        ]
    }
)
result = response.json()
print(result)
```

▼ curl

```
curl
```

```
https://api.cloudflare.com/client/v4/accounts/3-12b-it \
-X POST \
-H "Authorization: Bearer $CLOUDFLARE_AUTH_TOKEN" \
-d '{ "messages": [{ "role": "system", "content": "You are a helpful assistant." }, { "role": "user", "content": "Why is pizza so good" } ] }'
```

① OpenAI compatible endpoints

Workers AI also supports OpenAI compatible API endpoints for `/v1/chat/completions` and `/v1/embeddings`. For more details, refer to [Configurations](#).

Parameters

* indicates a required field

Input

• **object**

- **prompt** **string** required min 1

The input text prompt for the model to generate a response.

- **guided_json** **object**

JSON schema that should be fulfilled for the response.

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **max_tokens** **integer** default 256

The maximum number of tokens to generate in the response.

- temperature **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- top_p **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- top_k **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- seed **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- repetition_penalty **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

- ▶ 1 **object**

- `messages` **array** required

An array of message objects representing the conversation history.

- `items` **object**

- `role` **string**

The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- `content` **one of**

- `0` **string**

The content of the message as a string.

- 1 **array**

- items **object**

- type **string**

- Type of the content provided

- text **string**

- image_url **object**

- url **string**

- image uri with data (e.g.

- ...).

- HTTP URL will not be accepted

- 2 **object**

- type **string**

- Type of the content provided

- text **string**

- image_url **object**

- `url` **string**

image uri with data (e.g.

`...`).

HTTP URL will not be accepted

- `functions` **array**

- `items` **object**

- `name` **string** required

- `code` **string** required

- `tools` **array**

A list of tools available for the assistant to use.

- `items` **one of**

- `0` **object**

- `name` **string** required

- The name of the tool. More descriptive the better.

- `description` **string** required

A brief description of what the tool does.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

■ 1 **object**

- **type** **string** required

Specifies the type of tool (e.g., 'function').

- **function** **object** required

Details of the function tool.

- **name** **string** required

The name of the function.

- **description** **string** required

A brief description of what the function does.

- **parameters** **object** required

Schema defining the parameters accepted by the function.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **guided_json** **object**

JSON schema that should be fulfilled for the response.

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **max_tokens** **integer** default 256

The maximum number of tokens to generate in the response.

- **temperature** **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- **top_p** **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

Output

- **0** **object**

- response **string** required

The generated text response from the model

- usage **object**

Usage statistics for the inference request

- prompt_tokens **number** 0

Total number of tokens in input

- completion_tokens **number** 0

Total number of tokens in output

- total_tokens **number** 0

Total number of input and output tokens

- tool_calls **array**

An array of tool calls requests made during the response generation

- items **object**

- arguments **object**

The arguments passed to be passed to the tool call request

- name **string**

The name of the tool to be called

- ▶ 1 **string**

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{  
  "type": "object",  
  "oneOf": [  
    {  
      "title":  
"Google_Gemma_3_12B_It_Prompt",  
      "properties": {  
        "prompt": {  
          "type": "string",  
          "minLength": 1,  
          "description": "The input  
text prompt for the model to generate a  
response."  
        },  
        "guided_json": {  
          "type": "object",  
          "description": "JSON  
schema that should be fulfilled for the  
response."  
        },  
        "raw": {  
          "type": "boolean",  
          "default": false,  

```


`"description": "If true, a chat template is not applied and you must adhere to the specific model's expected formatting."`

`},`

`"stream": {`

`"type": "boolean",`

`"default": false,`

`"description": "If true,`

the response will be streamed back

incrementally using SSE, Server Sent Events."

`},`

`"max_tokens": {`

`"type": "integer",`

`"default": 256,`

`"description": "The`

maximum number of tokens to generate in the

response."

`},`

`"temperature": {`

`"type": "number",`

`"default": 0.6,`

`"minimum": 0,`

`"maximum": 5,`

```
    "description": "Controls  
the randomness of the output; higher values  
produce more random results."
```

```
  },
```

```
  "top_p": {
```

```
    "type": "number",
```

```
    "minimum": 0,
```

```
    "maximum": 2,
```

```
    "description": "Adjusts  
the creativity of the AI's responses by  
controlling how many possible words it  
considers. Lower values make outputs more  
predictable; higher values allow for more  
varied and creative responses."
```

```
  },
```

```
  "top_k": {
```

```
    "type": "integer",
```

```
    "minimum": 1,
```

```
    "maximum": 50,
```

```
    "description": "Limits the  
AI to choose from the top 'k' most probable  
words. Lower values make responses more  
focused; higher values introduce more variety  
and potential surprises."
```

```
  },
```

```
"seed": {  
  "type": "integer",  
  "minimum": 1,  
  "maximum": 9999999999,  
  "description": "Random  
seed for reproducibility of the generation."  
},  
"repetition_penalty": {  
  "type": "number",  
  "minimum": 0,  
  "maximum": 2,  
  "description": "Penalty  
for repeated tokens; higher values discourage  
repetition."  
},  
"frequency_penalty": {  
  "type": "number",  
  "minimum": 0,  
  "maximum": 2,  
  "description": "Decreases  
the likelihood of the model repeating the same  
lines verbatim."  
},  
"presence_penalty": {  
  "type": "number",
```

```
        "minimum": 0,
        "maximum": 2,
        "description": "Increases
the likelihood of the model introducing new
topics."
    }
},
"required": [
    "prompt"
]
},
{
    "title":
"Google_Gemma_3_12B_It_Messages",
    "properties": {
        "messages": {
            "type": "array",
            "description": "An array
of message objects representing the
conversation history.",
            "items": {
                "type": "object",
                "properties": {
                    "role": {
```

```
        "type":  
        "string",  
        "description":  
        "The role of the message sender (e.g., 'user',  
        'assistant', 'system', 'tool')."  
    },  
    "content": {  
        "oneOf": [  
            {  
  
        "type": "string",  
  
        "description": "The content of the message as  
        a string."  
            },  
            {  
  
        "type": "array",  
  
        "items": {  
  
        "type": "object",  
  
        "properties": {
```

```
"type": {  
  
  "type": "string",  
  
  "description": "Type of the content provided"  
  
},  
  
"text": {  
  
  "type": "string"  
  
},  
  
"image_url": {  
  
  "type": "object",  
  
  "properties": {  
  
    "url": {  
  
      "type": "string",
```

```
"pattern": "^data:",
```

```
"description": "image uri with data (e.g.  
...). HTTP URL will  
not be accepted"
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
},
```

```
{
```

```
"type": "object",
```

```
"properties": {
```

```
"type": {
```

```
"type": "string",
```

```
"description": "Type of the content provided"
},

"text": {

"type": "string"
},

"image_url": {

"type": "object",

"properties": {

"url": {

"type": "string",

"pattern": "^data:*",

"description": "image uri with data (e.g.
...). HTTP URL will
not be accepted"
```



```
}

}

}

}

}

}

}

}],
"functions": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "name": {
        "type":
"string"

      },
      "code": {
        "type":
"string"

      }
    }
  }
}
```

```
    },  
    "required": [  
        "name",  
        "code"  
    ]  
  }  
},  
"tools": {  
  "type": "array",  
  "description": "A list of  
tools available for the assistant to use.",  
  "items": {  
    "type": "object",  
    "oneOf": [  
      {  
        "properties":  
  
        "name": {  
  
          "type": "string",  
  
          "description": "The name of the tool. More  
descriptive the better."  
        },  
      },  
    ],  
  },  
}
```

```
"description": {  
  
  "type": "string",  
  
  "description": "A brief description of what  
the tool does."  
  
},  
  
"parameters": {  
  
  "type": "object",  
  
  "description": "Schema defining the parameters  
accepted by the tool.",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "The type of the parameters  
object (usually 'object')."  
  
    },  
  
  },  
  
}
```

```
"required": {  
  
  "type": "array",  
  
  "description": "List of required parameter  
names.",  
  
  "items": {  
  
    "type": "string"  
  }  
},  
  
  "properties": {  
  
    "type": "object",  
  
    "description": "Definitions of each  
parameter.",  
  
    "additionalProperties": {  
  
      "type": "object",
```

```
"properties": {  
  
  "type": {  
  
    "type": "string",  
  
    "description": "The data type of the  
parameter."  
  
  },  
  
  "description": {  
  
    "type": "string",  
  
    "description": "A description of the expected  
parameter."  
  
  },  
  
  "required": [  

```

```
"type",  
  
"description"  
  
]  
  
}  
  
    },  
  
"required": [  
  
"type",  
  
"properties"  
  
    ]  
    }  
  },  
  "required": [  
    "name",  
  
    "description",  
  
    "parameters"
```

```
        ]
    },
    {
        "properties":
    {
        "type": {

"type": "string",

"description": "Specifies the type of tool
(e.g., 'function')."

        },

"function": {

"type": "object",

"description": "Details of the function
tool.",

"properties": {

"name": {

"type": "string",
```

```
"description": "The name of the function."  
    },
```

```
"description": {
```

```
"type": "string",
```

```
"description": "A brief description of what  
the function does."  
    },
```

```
"parameters": {
```

```
"type": "object",
```

```
"description": "Schema defining the parameters  
accepted by the function.",
```

```
"properties": {
```

```
"type": {
```

```
"type": "string",
```



```
"description": "The type of the parameters  
object (usually 'object')."
```

```
},
```

```
"required": {
```

```
"type": "array",
```

```
"description": "List of required parameter  
names.",
```

```
"items": {
```

```
"type": "string"
```

```
}
```

```
},
```

```
"properties": {
```

```
"type": "object",
```

```
"description": "Definitions of each  
parameter.",  
  
"additionalProperties": {  
  
  "type": "object",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "The data type of the  
parameter."  
  
    },  
  
    "description": {  
  
      "type": "string",  
  
      "description": "A description of the expected  
parameter."
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"description"
```

```
]
```

```
}
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"properties"
```

```
]
    },
    "required": [
        "name",
        "description",
        "parameters"
    ]
},
"required": [
    "type",
    "function"
]
}
],
},
"guided_json": {
    "type": "object",
```

```
      "description": "JSON
schema that should be fulfilled for the
response."
    },
    "raw": {
      "type": "boolean",
      "default": false,
      "description": "If true, a
chat template is not applied and you must
adhere to the specific model's expected
formatting."
    },
    "stream": {
      "type": "boolean",
      "default": false,
      "description": "If true,
the response will be streamed back
incrementally using SSE, Server Sent Events."
    },
    "max_tokens": {
      "type": "integer",
      "default": 256,
      "description": "The
maximum number of tokens to generate in the
response."
    }
  }
}
```

```
},  
  "temperature": {  
    "type": "number",  
    "default": 0.6,  
    "minimum": 0,  
    "maximum": 5,  
    "description": "Controls  
the randomness of the output; higher values  
produce more random results."  
  },  
  "top_p": {  
    "type": "number",  
    "minimum": 0,  
    "maximum": 2,  
    "description": "Adjusts  
the creativity of the AI's responses by  
controlling how many possible words it  
considers. Lower values make outputs more  
predictable; higher values allow for more  
varied and creative responses."  
  },  
  "top_k": {  
    "type": "integer",  
    "minimum": 1,  
    "maximum": 50,
```

`"description": "Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises."`

`},`

`"seed": {`

`"type": "integer",`

`"minimum": 1,`

`"maximum": 9999999999,`

`"description": "Random seed for reproducibility of the generation."`

`},`

`"repetition_penalty": {`

`"type": "number",`

`"minimum": 0,`

`"maximum": 2,`

`"description": "Penalty for repeated tokens; higher values discourage repetition."`

`},`

`"frequency_penalty": {`

`"type": "number",`

`"minimum": 0,`

`"maximum": 2,`

```
      "description": "Decreases  
the likelihood of the model repeating the same  
lines verbatim."  
    },  
    "presence_penalty": {  
      "type": "number",  
      "minimum": 0,  
      "maximum": 2,  
      "description": "Increases  
the likelihood of the model introducing new  
topics."  
    }  
  },  
  "required": [  
    "messages"  
  ]  
}
```

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